$$\theta = 30^\circ \qquad C = \cos 30^\circ = \frac{\sqrt{3}}{2} \qquad S = \sin 30^\circ = \frac{1}{2}$$

$$[k] = \frac{(2)(30 \times 10^6)}{60} \begin{bmatrix} \frac{3}{4} & \frac{\sqrt{3}}{4} & \frac{-3}{4} & \frac{-\sqrt{3}}{4} \\ & \frac{1}{4} & \frac{-\sqrt{3}}{4} & \frac{-1}{4} \\ & & \frac{3}{4} & \frac{\sqrt{3}}{4} \\ \text{Symmetry} & & & \frac{1}{4} \end{bmatrix} \frac{\text{lb}}{\text{in.}} \tag{3.4.27}$$

Simplifying Eq. (3.4.27), we have

$$[k] = 10^6 \begin{bmatrix} 0.75 & 0.433 & -0.75 & -0.433 \\ & 0.25 & -0.433 & -0.25 \\ & & 0.75 & 0.433 \\ \text{Symmetry} & & & 0.25 \end{bmatrix} \frac{\text{lb}}{\text{in.}} \tag{3.4.28}$$

3.5 Computation of Stress for a Bar in the $x-y$ Plane

We will now consider the determination of the stress in a bar element. For a bar, the local forces are related to the local displacements by Eq. (3.4.1) or Eq. (3.4.17). This equation is repeated here for convenience.

$$\begin{Bmatrix} f'_{1x} \\ f'_{2x} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u'_1 \\ u'_2 \end{Bmatrix} \tag{3.5.1}$$

The usual definition of axial tensile stress is axial force divided by cross-sectional area. Therefore, axial stress is

$$\sigma = \frac{f'_{2x}}{A} \tag{3.5.2}$$

where f'_{2x} is used because it is the axial force that pulls on the bar as shown in Figure 3–11. By Eq. (3.5.1),

$$f'_{2x} = \frac{AE}{L} [-1 \quad 1] \begin{Bmatrix} u'_1 \\ u'_2 \end{Bmatrix} \tag{3.5.3}$$

Therefore, combining Eqs. (3.5.2) and (3.5.3) yields

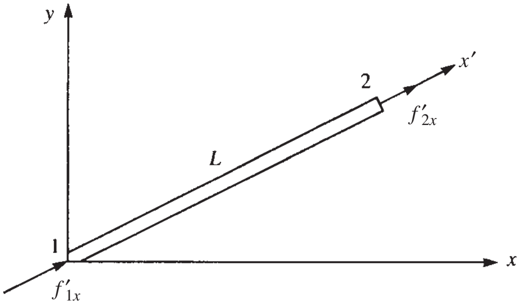

Figure 3–11 Basic bar element with positive nodal forces

$$\{\sigma\} = \frac{E}{L}\begin{bmatrix}-1 & 1\end{bmatrix}\{d'\} \tag{3.5.4}$$

Now, using Eq. (3.4.7), we obtain

$$\{\sigma\} = \frac{E}{L}\begin{bmatrix}-1 & 1\end{bmatrix}[T^*]\{d\} \tag{3.5.5}$$

Equation (3.5.5) can be expressed in simpler form as

$$\{\sigma\} = [C']\{d\} \tag{3.5.6}$$

where, when we use Eq. (3.4.8) for $[T^*]$,

$$[C'] = \frac{E}{L}\begin{bmatrix}-1 & 1\end{bmatrix}\begin{bmatrix}C & S & 0 & 0\\ 0 & 0 & C & S\end{bmatrix} \tag{3.5.7}$$

After multiplying the matrices in Eq. (3.5.7), we have

$$[C'] = \frac{E}{L}[-C \quad -S \quad C \quad S] \tag{3.5.8}$$

EXAMPLE 3.4

For the bar shown in Figure 3–12, determine the axial stress. Let $A = 4 \times 10^{-4}$ m^2, $E = 210$ GPa, and $L = 2$ m, and let the angle between x and x' be 60°. Assume the global displacements have been previously determined to be $u_1 = 0.25$ mm, $v_1 = 0.0$, $u_2 = 0.50$ mm, and $v_2 = 0.75$ mm.

SOLUTION:

We can use Eq. (3.5.6) to evaluate the axial stress. Therefore, we first calculate $[C']$ from Eq. (3.5.8) as

$$[C'] = \frac{210 \times 10^6 \text{ kN/m}^2}{2 \text{ m}}\begin{bmatrix}\frac{-1}{2} & \frac{-\sqrt{3}}{2} & \frac{1}{2} & \frac{\sqrt{3}}{2}\end{bmatrix} \tag{3.5.9}$$